

1. What is the primary difference between Proof of Work (PoW) and Proof of Stake (PoS)?
 - a) PoW is more energy-efficient than PoS
 - b) PoW validators are chosen based on computational power (work), while PoS validators are chosen based on the number of coins they hold (stake)
 - c) PoW is used for private blockchains and PoS is for public blockchains
 - d) Only PoW allows for the creation of smart contracts

Answer: b

2. What is the primary purpose of a Merkle Tree in a block?
 - a) To organize user identities
 - b) To calculate block reward for miners
 - c) To efficiently and securely verify the integrity of a large set of transactions
 - d) To rank miners based on their hashing power

Answer: c

3. A blockchain governed by a group of pre-selected organizations is called a:
 - a) Public blockchain
 - b) Private blockchain
 - c) Consortium blockchain
 - d) Hybrid blockchain

Answer: c

4. ERC-20 is a technical standard for tokens on which blockchain?
 - a) Bitcoin
 - b) Cardano
 - c) Solana
 - d) Ethereum

Answer: d

5. What is the main goal of Layer 2 solutions like the Lightning Network or Polygon?
 - a) To improve blockchain security
 - b) To increase decentralization
 - c) To improve scalability by handling transactions off the main chain
 - d) To make blockchains anonymous

Answer: c

6. The "Blockchain Trilemma" refers to the difficulty of achieving which three properties simultaneously?
 - a) Speed, Cost, and User Experience
 - b) Anonymity, Transparency, and Immutability
 - c) Decentralization, Security, and Scalability
 - d) Hardware, Software, and Network

Answer: c

7. Which consensus algorithm does Bitcoin use?
 - a) Proof of Stake

- b) Proof of Work
- c) Proof of Authority
- d) Proof of Burn

Answer: b

8. What is the size of a Bitcoin block?

- a) 512 MB
- b) 1 MB
- c) 2 MB
- d) 10 MB

Answer: b

9. Which of the following supports NFT creation?

- a) Ethereum
- b) Ripple
- c) Bitcoin
- d) Stellar

Answer: a

10. Which of the following is a Layer 2 solution for Ethereum?

- a) Polygon
- b) Solana
- c) Avalanche
- d) Cardano

Answer: a

11. Which programming language is used for Ethereum smart contracts?

- a) Python
- b) Solidity
- c) C++
- d) Go

Answer: b

12. Which of the following blockchains uses “Delegated Proof of Stake”?

- a) EOS
- b) Bitcoin
- c) Ethereum
- d) Hyperledger

Answer: a

13. The Ethereum Virtual Machine (EVM) executes:

- a) Bytecode of smart contracts
- b) JavaScript code
- c) Hashing functions only
- d) APIs

Answer: a

14. "Stablecoins" derive their value from:
- a) Central bank reserves or real-world assets
 - b) Cryptographic randomness
 - c) Fixed number generation
 - d) Proof validation

Answer: a

15. What is "cold storage" in crypto?
- a) Offline storage of digital assets
 - b) Encrypted transaction
 - c) Trading algorithm
 - d) Backup in cloud

Answer: a

16. DApps are primarily built on top of:
- a) Central banks
 - b) Blockchain protocols
 - c) Firewalls
 - d) Web browsers

Answer: b

17. What is "gas" in Ethereum?
- a) A fee paid to execute transactions and smart contracts
 - b) A reward given to miners
 - c) An ERC-20 token
 - d) A decentralized storage service

Answer: a

18. Which blockchain uses the UTXO (Unspent Transaction Output) model?
- a) Ethereum
 - b) Bitcoin
 - c) Cardano only
 - d) Solana

Answer: b

19. What does a "Genesis Block" refer to?
- a) The first block in a blockchain network
 - b) The last block before halving
 - c) A block used for mining rewards
 - d) The transaction index block

Answer: a

20. In smart contracts, what happens when a condition is not met?
- a) The contract automatically fails and reverts changes
 - b) The network deletes the contract
 - c) The user gets a partial refund

d) The miner modifies the contract manually

Answer: a

21. What is the primary role of a consensus mechanism?

- a) To verify and agree on the validity of transactions
- b) To encrypt private keys
- c) To manage token circulation
- d) To build APIs

Answer: a

22. Which type of blockchain allows both public and private access?

- a) Consortium blockchain
- b) Private blockchain
- c) Public blockchain
- d) Permissionless blockchain

Answer: a

23. What is the purpose of the “Nonce” in mining?

- a) A random number used to find a valid hash for a block
- b) A transaction verification tool
- c) A smart contract validator
- d) A wallet identifier

Answer: a

24. What is “Tokenomics”?

- a) The study of token supply, demand, and economic structure in a blockchain ecosystem
- b) The process of encrypting tokens
- c) Mining algorithm efficiency
- d) DeFi network fees

Answer: a

25. What happens when a blockchain network undergoes a “hard fork”?

- a) The blockchain splits into two separate chains
- b) All nodes are reset
- c) Mining stops temporarily
- d) The token value is halved

Answer: a